DICHOTOMY FOR IRREDUCIBLE POLYNOMIALS: COPRIME DERIVATIVE OR P-POWER SUBSTITUTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This lemma is the engine of the purely inseparable theory: P(x) = Q(x^q) is what makes alpha^q separable in lemma-separable-first.

Editorial note. The label lemma-irreducible-polynomials names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Polynomial algebra over a field: k[x] is a PID so (P, P') = (R); the derivative of a polynomial in characteristic p kills exactly the terms of degree divisible by p

Lemma 0.1. *Let F be a field. Let $P \in F[x]$ be an irreducible polynomial over F. Let $P' = dP/dx$ be the derivative of P with respect to x. Then one of the following two cases happens*

(1) *P and P' are relatively prime, or*
(2) *P' is the zero polynomial.*

The second case can only happen if F has characteristic $p > 0$. In this case $P(x) = Q(x^q)$ where $q = p^f$ is a power of p and $Q \in F[x]$ is an irreducible polynomial such that Q and Q' are relatively prime.

Proof. Note that P' has degree $< \deg(P)$. Hence if P and P' are not relatively prime, then $(P, P') = (R)$ where R is a polynomial of degree $< \deg(P)$ contradicting the irreducibility of P. This proves we have the dichotomy between (1) and (2).

Assume we are in case (2) and $P = a_d x^d + \ldots + a_0$. Then $P' = d a_d x^{d-1} + \ldots + a_1$. In characteristic 0 we see that this forces $a_d, \ldots, a_1 = 0$ which would mean P is constant a contradiction. Thus we conclude that the characteristic p is positive. In this case the condition $P' = 0$ forces $a_i = 0$ whenever p does not divide i. In other words, $P(x) = P_1(x^p)$ for some nonconstant polynomial P_1. Clearly, P_1 is irreducible as well. By induction on the degree we see that $P_1(x) = Q(x^q)$ as in the statement of the lemma, hence $P(x) = Q(x^{pq})$ and the lemma is proved. □

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